

CSTE 5131

Machine Learning

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Classification: Basic Concepts

- Classification problem overview
- General issues of classification DM
- Mining classification model by decision tree induction
- **Bayesian Classification**
- Other classification methods
- Summary

Bayesian Classification



- What are Bayesian classifiers?
 - Bayesian classifiers are statistical classifiers. They can predict class membership probabilities such as the probability that a given tuple belongs to a particular class.
- Bayesian classification is based on **Bayes' theorem**
 - Classify each new case by directly calculating the probable class based on the Bayes rule and the prior probabilities matching the case
- A simple Bayesian classifier known as the ***naive Bayesian classifier***
 - Naive Bayesian classifiers assume that the effect of an attribute value on a given class is independent of the values of the other attributes. This assumption is called ***class conditional independence***.



Bayes' Theorem

- Bayes Rule: (Thomas Bayes)
 - If you have a hypothesis **h**, and an evidence **e** which supports the hypothesis, then

$$P(h|e) = \frac{P(e|h) P(h)}{P(e)},$$

$$P(e) = P(e|h)P(h) + P(e|\sim h)P(\sim h) \quad \leftarrow \text{Law of total probability}$$

where,

$P(h|e)$: posterior probability of h conditioned on e

$P(h)$: prior probability of h

$P(e|h)$: posterior probability of e conditioned on h

$P(e)$: prior probability of e

Illustration of Bayes' Rule Application

- **Example of medical diagnosis**

E.g., h = "John has malaria", and e = "John has a high fever."

General knowledge:

1. $P(h)$: probability that a person has malaria in a particular season and region.
2. $P(e | h)$: probability that a person has a high fever, given that he has malaria.
3. $P(e)$: probability that a person has a high fever.

Given the general knowledge: $P(h) = 0.0001$, $P(e | h) = 0.75$, $P(e | \sim h) = 0.14$.

$P(e) = (0.75)(0.0001) + (0.14)(0.9999) = 0.14006$, and

$$P(h | e) = P(e | h)P(h) / P(e) = (0.75)(0.0001) / 0.14006 = 0.0005354$$

On the other hand, if John did not have the fever:

$$P(h | \sim e) = P(\sim e | h)P(h) / P(\sim e) = (1 - 0.75)(0.0001) / (1 - 0.14006) = 0.0000029$$

Naïve Bayesian Classification

- Naïve Bayes Classifier (Instance based)

$$C_{NB} = \underset{j \in \{yes, no\}}{\operatorname{argmax}} P(c_j) \prod_{i=1}^n P(x_i | c_j)$$

- Explanation in the class

$$c_{NB} = \underset{j \in \{yes, no\}}{\operatorname{argmax}} P(c_j) \prod_{i=1}^n P(x_i | c_j)$$

Dataset

$$X = \{x_1, x_2, x_3, \dots, x_n\} \quad \{y\}$$

$x_1 \quad x_2 \quad x_3 \quad \dots \quad x_n$

$$P(h|e) = \frac{P(e|h) P(h)}{P(e)}$$

$\downarrow$

$$P(y|x_1, x_2, \dots, x_n) = \frac{P(x_1|y) P(x_2|y) \dots P(x_n|y) * P(y)}{P(x_1) P(x_2) \dots P(x_n)}$$

$$= \frac{P(y) \prod_{i=1}^n P(x_i|y)}{P(x_1) P(x_2) \dots P(x_n)}$$

$$P(y|x_1, x_2, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i|y)$$

$$y = \underset{y}{\operatorname{argmax}} P(y) \prod_{i=1}^n P(x_i|y)$$



Example: Naïve Bayesian Classification

- E.g., Given the data set below, classify the input instance (sunny, cool, high, strong, ?) for the concept PlayTennis

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	sunny	hot	high	weak	no
2	sunny	hot	high	strong	no
3	overcast	hot	high	weak	yes
4	rain	mild	high	weak	yes
5	rain	cool	normal	weak	yes
6	rain	cool	normal	strong	no
7	overcast	cool	normal	strong	yes
8	sunny	mild	high	weak	no
9	sunny	cool	normal	weak	yes
10	rain	mild	normal	weak	yes
11	sunny	mild	normal	strong	yes
12	overcast	mild	high	strong	yes
13	overcast	hot	normal	weak	yes
14	rain	mild	high	strong	no



Example: Naïve Bayesian Classification

- Apply Naïve Bayes classifier for the new instance:

(Sunny, cool, high, strong, ?)

Apply Naïve Bayes classifier:

$$C_{NB} = \underset{j \in \{yes, no\}}{argmax} P(c_j) \prod_{i=1}^n P(x_i | c_j)$$

- $c_{yes} = P(\text{Yes}) * P(\text{Outlook=sunny} | \text{yes}) * P(\text{Temp=cool} | \text{yes}) * P(\text{Hum=high} | \text{yes}) * P(\text{Wind=strong} | \text{yes})$
- $c_{no} = P(\text{No}) * P(\text{Outlook=sunny} | \text{no}) * P(\text{Temp=cool} | \text{no}) * P(\text{Hum=high} | \text{no}) * P(\text{Wind=strong} | \text{no})$



Example: Naïve Bayesian Classification

- How to calculate prior and conditional probabilities?

E.g., $P(\text{Outlook}=\text{sunny} \mid \text{Yes}) = 2/9$

$P(\text{Outlook}=\text{sunny} \mid \text{No}) = 3/5$

The prior knowledge in the training data about the target "Play-tennis":

Counts	Outlook		Temperature		Humidity		Wind		Play-tennis	
	yes	no	yes	no	yes	no	yes	no	yes	no
	sunny: 2 overca: 4 rainy: 3	3 0 2	hot: 2 mild: 4 cool: 3	2 2 1	high: 3 normal: 6	4 1	week: 6 strong: 3	2 3	9	5
Probabilities	sunny: 2/9 overca: 4/9 rainy: 3/9	3/5 0/5 2/5	hot: 2/9 mild: 4/9 cool: 3/9	2/5 2/5 1/5	high: 3/9 normal: 6/9	4/5 1/5	week: 6/9 strong: 3/9	2/5 3/5	9/14	5/14



Example: Naïve Bayesian Classification

- Prior and conditional probabilities for the new instance:

(sunny, cool, high, strong, ?)

$$\begin{aligned}P(c_{yes}) &= 9/14 = .64, \\P(\text{sunny} | c_{yes}) &= 2/9 = .22, \\P(\text{cool} | c_{yes}) &= 3/9 = .33, \\P(\text{high} | c_{yes}) &= 3/9 = .33, \\P(\text{strong} | c_{yes}) &= 3/9 = .33,\end{aligned}$$

$$\begin{aligned}P(c_{no}) &= 5/14 = .36 \\P(\text{sunny} | c_{no}) &= 3/5 = .60 \\P(\text{cool} | c_{no}) &= 1/5 = .20 \\P(\text{high} | c_{no}) &= 4/5 = .80 \\P(\text{strong} | c_{no}) &= 3/5 = .60\end{aligned}$$

Example: Naïve Bayesian Classification

Calculate Posterior Probabilities:

$$\begin{aligned} c_{\text{yes}} &= P(c_{\text{yes}}) * P(\text{sunny}|\text{yes}) * P(\text{cool}|\text{yes}) * P(\text{high}|\text{yes}) * P(\text{strong}|\text{yes}) \\ &= .64 * .22 * .33 * .33 * .33 = \mathbf{.0053} \end{aligned}$$

$$\begin{aligned} c_{\text{no}} &= P(c_{\text{no}}) * P(\text{sunny}|\text{no}) * P(\text{cool}|\text{no}) * P(\text{high}|\text{no}) * P(\text{strong}|\text{no}) \\ &= .36 * .6 * .2 * .8 * .6 = \mathbf{.0206} \end{aligned}$$

$$\mathbf{c_{yes} = .0053}$$

$$\mathbf{c_{no} = .0206}$$

Thus, the Naive Bayes classifier assigns the target value **PlayTennis = no.**



Example: Naïve Bayesian Classification

Classification for the input data:

(Outlook=sunny, Temp=cool, Hum=high, Wind=strong, Pay-tennis=?)

- The new input day:

Outlook	Temperature	Humidity	Wind	Play-tennis
sunny	cool	high	strong	?

- **The inference:**

likelihood of yes = $9/14 * 2/9 * 3/9 * 3/9 * 3/9 = 0.0053$

likelihood of no = $5/14 * 3/5 * 1/5 * 4/5 * 3/5 = 0.0206$

The numbers can be turned into probabilities by normalizing them so that they sum to 1

$$\text{Probability of Yes} = \frac{0.0053}{0.0053 + 0.0206} = 20.5\%$$

$$\text{Probability of No} = \frac{0.0206}{0.0053 + 0.0206} = 79.5\%$$

Predication: (sunny, cool, high, strong, No)



Avoid Computing Probability Values of Zero

- Problem with probability values of zero?
- Solution: **Laplace Smoothing (add-one smoothing)** (See also Section 8.3.2)
 - From previous example,

$$P(\text{sunny} | C_{\text{yes}}) = \frac{2}{9}$$

$$P(\text{sunny} | C_{\text{yes}}) = \frac{2 + \alpha}{9 + \alpha K}$$

Smoothing parameter

- For text classification, the denominator will add $|V|$ (number of unique words in vocabulary)



Observations on Naïve Bayesian Classifier

- Strength:
 - Easy to implement
 - Robust to isolated noise points
 - Handle missing values by ignoring the instance during probability estimate calculations
 - Robust to irrelevant attributes, and high dimensional data
 - Good results obtained in some application domains, e.g. text classification
- Limitations:
 - Assumption: attributes independent, therefore may loss accuracy
 - Practically, dependencies exist among variables
 - E.g., population data: age, education, wage, etc.
 - E.g., medical diagnoses data (symptoms): fever, cough, etc.
 - Dependencies among factors cannot be modeled by Naïve Bayesian Classifier



Naïve Bayes (NB) for Text Classification

- What is text classification?
 - Consider a classification problem in which the **instances are text documents**:
 - Training data
 - Target classes
 - Prediction: label an unseen text to one of the classes
- Application examples:
 - Spam filtering
 - Text document categorization (place e-articles into categories), such as news articles, tweets, etc.
 - Classify survey comments into positive or negative class



Dataset: Structured vs. Unstructured

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	sunny	hot	high	weak	no
2	sunny	hot	high	strong	no
3	overcast	hot	high	weak	yes
4	rain	mild	high	weak	yes
5	rain	cool	normal	weak	yes
6	rain	cool	normal	strong	no
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11	sunny	mild	normal	strong	yes
12	overcast	mild	high	strong	yes
13	overcast	hot	normal	weak	yes
14	rain	mild	high	strong	no

Structured:

- Objects
- Attributes
- Target values



Dataset: Structured vs. Unstructured

Unstructured Text Data -> Structured Data

"Our approach to representing arbitrary text documents is disturbingly simple: given a text document, such as this paragraph, we define an attribute for each word position in the document and define the value of that attribute to be the English word found in that position. Thus, the current paragraph would be described by 111 attribute values, corresponding to the 111 word positions. The value of the first attribute is the word "Our," the value of the second attribute is the word "approach," and so on. Notice that long text documents will require a larger number of attributes than short documents. As we shall see, this will not cause us any trouble."

How to make text be structured for classification



- Text documents are unstructured data
 - In comparing with structured data, such as formatted data in a table
 - o With fixed number of attributes, and well-defined domain for each attribute, etc.
 - o What about text documents? -Structure, object unit, etc.
- How to automatically process large quantities of unstructured text data?
 - People are good at processing small quantities of unstructured text (We have intelligence)
 - Computers are good at processing large quantities of structured text



Main ideas for text classification

- Make text data be structured
 - We need to find ways to structure text so that it can be understood by computer:
 - o Treat each text document as an object
 - o Treat word position as attribute
 - o Treat words as domain values
- Apply a classification method which can handle high dimension data, such as **NB method**
 - How to represent text in terms of attributes?
 - How to calculate prior probabilities for using NB?



Example (Naïve Bayes): Text Classification

- Example and discussion in the class

Sentence 1 : The movie was good
Sent. 2 : The movie was bad
Sent. 3 : Movie is bad

	x_1	x_2	x_3	x_4	P/N
	The	movie	good	bad	
1.	1	1	1	0	1
2.	1	1	0	1	0
3.	0	1	0	1	0
4.	0	1	1	0	1
5.	0	0	0 1	1	0

$$P(Y=Yes) = 2/5$$
$$P(Y=N) = 3/5$$

$$P(y|x_1, x_2, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i|y)$$

$$P(y=Yes|x_1, x_2, \dots, x_n) \propto P(y=Yes) * P(x_1|y=Yes) * \dots * P(x_n|y=Yes)$$



Example (Naïve Bayes): Text Classification

- Document Preprocessing:
 - Preprocessing involves removal of punctuations, numbers, white spaces, stop words, etc.
 - stemming (assault, assaults, assaulting, assaulted all can be stemmed to the word "assault")
- Document Representation:
 - Term-Document Matrix (TDM of Bag-of-words (BOW))
 - Document Embedding

Homework!



- Read class note, and chapter 8 (mainly Section 8.1, 8.2, 8.3)

Thank you

